

CURRENT POSITION	Assistant Professor - Marquette University , Milwaukee, Wisconsin, USA Department of Mathematical and Statistical Sciences Start Date: August 2020	
WEB	https://gregongie.github.io Google Scholar	
RESEARCH INTERESTS	Mathematically principled approaches to machine learning in computational imaging, including image reconstruction in X-ray computed tomography and magnetic resonance imaging. Mathematical foundations of deep learning, with emphasis on function space perspectives and implicit bias of neural network training.	
DEGREES	University of Iowa , Iowa City, Iowa, USA Ph.D., Applied Mathematical and Computational Sciences, July 2016 • Thesis Topic: <i>Off-the-grid Compressive Imaging</i> • Advisor: Mathews Jacob, Ph.D. M.S., Mathematics, Aug 2011 Coe College , Cedar Rapids, Iowa, USA B.S., Mathematics and Physics, May 2008	Aug 2008 – July 2016
PREVIOUS POSITIONS	Postdoctoral Scholar - University of Chicago Committee on Computational and Applied Mathematics Supervisor: Rebecca Willett, Ph.D.	Aug 2018 – July 2020
	Postdoctoral Scholar - University of Michigan Department of Electrical Engineering and Computer Science Supervisors: Laura Balzano, Ph.D & Jeff Fessler, Ph.D.	Sept 2016 – July 2018

Research

Since my start as an assistant professor at Marquette University in 2020, I have built up a small but active research group focused on mathematically principled approaches to **machine learning in computational imaging**. From 2022-2025, my PhD students were funded under a National Science Foundation grant related to this topic. I have also submitted a follow-up NSF proposal as Co-PI (currently under review). Additionally, I have on-going research collaborations with national and international groups focused on the **mathematical foundations of deep learning**.

RESEARCH FUNDING	• National Science Foundation CISE Research Initiation Initiative Grant 2022-2025 <i>Coordinate-based Neural Networks for Inverse Problems in Computational Imaging</i>. Awarded March 2022. Total award: \$173,184 USD. Supported 2 graduate students (Megan Murphy, 2 academic years; Mahrokh Najaf, 1 academic year + 1 summer). • Summer Faculty Fellowship Award, Marquette University. 2025 <i>Enhanced Denoising of Breast CT Images Using Machine Learning</i>. Award: \$5,500.
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AWARDS	• Way Klinger Early Career Award, Marquette University	2023
	• D.C. Spriestersbach Outstanding Dissertation Prize in the Mathematical, Physical Sciences and Engineering, University of Iowa.	2018
	• Small Groups funding at the Alan Turing Institute “Theoretical and computational aspects of super-resolution in higher dimensions,” with A. Eftekhari, J. Tanner, and H. Tyagi.	2017
	• Best Student Paper Award: “Super-resolution MRI using finite rate of innovation curves,” IEEE International Symposium on Biomedical Imaging.	2015
PROFESSIONAL MEMBERSHIPS	• Society of Photo-Optical Instrumentation Engineers (SPIE)	2020–present
	• Society for Industrial and Applied Mathematics (SIAM)	2014–present
	• Institute of Electrical and Electronics Engineers (IEEE)	2013–present
PROFESSIONAL SERVICE	• Associate Editor of IEEE Transactions on Computational Imaging	2025-present
	• Member of the IEEE Computational Imaging Technical Committee	2021-present
	• Chair of Subcommittee on Special Sessions and Issues	2024-2025
	• U.S. National Science Foundation (NSF) Grant Review Panelist	2022 & 2023
	• Area Chair for IEEE International Conference on Image Processing (ICIP)	2026
	• Area Chair for the Conference on Parsimony in Learning (CPAL),	2023 & 2025
	• Area Chair for IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP) in the Biomedical Signal Processing Track.	2021 & 2022
	• Technical paper reviewer	
	• Applied mathematics journals	
	• Applied and Computational Harmonic Analysis	
	• SIAM Journal on Mathematics of Data Science	
	• SIAM Journal on Imaging Science	
	• AI/ML journals and conferences	
	• Neural Information Processing Systems (NeurIPS)	
	• Conference on Learning Theory (COLT)	
	• International Conference on Artificial Intelligence and Statistics (AISTATS)	
	• Journal of Machine Learning Research (JMLR)	
	• Workshop on Learning for Computational Imaging at ICCV	
	• Medical imaging/computational imaging journals	
	• IEEE Trans. on Medical Imaging	
	• IEEE Trans. on Computational Imaging	
	• Magnetic Resonance in Medicine	
	• Medical Physics	
	• Signal processing journals	
	• IEEE Trans. Signal Processing	
	• IEEE Trans. Pattern Analysis and Machine Intelligence	
	• IEEE Selected Topics in Signal Processing	
	• IEEE Signal Processing Letters	

CONFERENCE ORGANIZATION	• Special session: “Machine Learning Methods for Inverse Problems in Biomedical Imaging”. Asilomar Conference on Signals, Systems, and Computers, 2024. • Minisymposium: “Learning Functions with Low-Dimensional Structure using Neural Networks”. SIAM Conference on the Mathematics of Data Science, 2024. Co-organized with R. Willett & S. Parkinson. • Minisymposium: “Implicit Neural Representations for Inverse Problems in Imaging”. SIAM Conference on Imaging Science, 2024. • Minisymposium: “Implicit bias and regularized in overparametrized deep networks”. SIAM Conference on Mathematics of Data Science, 2022. Co-organized with R. Willett. • Special session: “Model-based Deep Learning for Inverse Problems in Imaging”. Asilomar Conference on Signals, Systems, and Computers, 2021. • Special session: “Smart Imaging Systems”. IEEE International Symposium on Biomedical Imaging (ISBI), 2018. Co-organized with S. Ravishankar & J. Fessler. • Special session: “Structured and Covariance Matrix Recovery”. Asilomar Conference on Signals, Systems, and Computers, 2017. Co-organized with L. Balzano.																								
RECENT INVITED TALKS	<table> <tr> <td>• “A Function Space View of Neural Networks.” ECE Department Colloquium, University of Delaware, Newark, DE.</td><td>Mar 2026</td></tr> <tr> <td>• “A Function Space View of Neural Networks.” Math Department Colloquium, University of Wisconsin-Milwaukee, Milwaukee, WI.</td><td>Oct 2025</td></tr> <tr> <td>• “A Function Space View of Norm-Constrained Deep Neural Networks.” SIAM Annual Meeting, Montreal, Canada.</td><td>July 2025</td></tr> <tr> <td>• “A Function-Space Tour of Data Science.” Conference on Parsimony and Learning (CPAL), Stanford, CA. (3 hour tutorial, co-presented with R. Parhi, UC San Diego).</td><td>Mar 2025</td></tr> <tr> <td>• “Direct optimization of signal detection metrics in learning-based CT image restoration.” SPIE Medical Imaging, San Diego, CA.</td><td>Feb 2025</td></tr> <tr> <td>• “Machine learning for enhanced image reconstruction in medical imaging.” MSSC Department Colloquium, Marquette University, Milwaukee, WI.</td><td>Oct 2024</td></tr> <tr> <td>• “Sampling Theorems for Implicit Neural Representations.” SIAM Conference on Imaging Science, Atlanta, GA.</td><td>May 2024</td></tr> <tr> <td>• “A function space view of infinite-width neural networks.” North Carolina State University, CAM seminar. [Virtual]</td><td>Nov 2023</td></tr> <tr> <td>• “Learned Reconstruction in Medical Imaging: Challenges and Opportunities.” University of Wisconsin-Madison, ECE Seminar.</td><td>Mar 2023</td></tr> <tr> <td>• “Exact recovery of coordinate-based neural network image representations from Fourier samples.” Asilomar Conference on Signals, Systems, and Computers. [Virtual]</td><td>Nov 2022</td></tr> <tr> <td>• “The role of linear layers in nonlinear interpolating networks.” SIAM Conference on the Mathematics of Data Science, San Diego, CA.</td><td>Sept 2022</td></tr> <tr> <td>• “Model-based deep learning for image reconstruction.” Medical College of Wisconsin, Milwaukee, WI. [Virtual]</td><td>Aug 2022</td></tr> </table>	• “A Function Space View of Neural Networks.” ECE Department Colloquium, University of Delaware, Newark, DE.	Mar 2026	• “A Function Space View of Neural Networks.” Math Department Colloquium, University of Wisconsin-Milwaukee, Milwaukee, WI.	Oct 2025	• “A Function Space View of Norm-Constrained Deep Neural Networks.” SIAM Annual Meeting, Montreal, Canada.	July 2025	• “A Function-Space Tour of Data Science.” Conference on Parsimony and Learning (CPAL), Stanford, CA. (3 hour tutorial, co-presented with R. Parhi, UC San Diego).	Mar 2025	• “Direct optimization of signal detection metrics in learning-based CT image restoration.” SPIE Medical Imaging, San Diego, CA.	Feb 2025	• “Machine learning for enhanced image reconstruction in medical imaging.” MSSC Department Colloquium, Marquette University, Milwaukee, WI.	Oct 2024	• “Sampling Theorems for Implicit Neural Representations.” SIAM Conference on Imaging Science, Atlanta, GA.	May 2024	• “A function space view of infinite-width neural networks.” North Carolina State University, CAM seminar. [Virtual]	Nov 2023	• “Learned Reconstruction in Medical Imaging: Challenges and Opportunities.” University of Wisconsin-Madison, ECE Seminar.	Mar 2023	• “Exact recovery of coordinate-based neural network image representations from Fourier samples.” Asilomar Conference on Signals, Systems, and Computers. [Virtual]	Nov 2022	• “The role of linear layers in nonlinear interpolating networks.” SIAM Conference on the Mathematics of Data Science, San Diego, CA.	Sept 2022	• “Model-based deep learning for image reconstruction.” Medical College of Wisconsin, Milwaukee, WI. [Virtual]	Aug 2022
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- “A function space view of infinite-width neural networks.” April 2022
Johns Hopkins MINDS/CIS Seminar. Baltimore, MD.
- “Optimizing model observer performance in learning-based April 2022
CT reconstruction.” U. Chicago Medical Physics Seminar. Chicago, IL.
- “Sketching Methods for Nonlinear Matrix Completion.” Nov 2021
Asilomar Conference on Signals, Systems, and Computers. [Virtual]
- “Learning to solve inverse problems in computational imaging” Oct 2021
Mitsubishi Electric Research Labs (MERL). [Virtual]

Advising and Mentoring

Below is a summary of my involvement in the Computational Mathematical and Statistical Sciences (CMPS) PhD and Master’s Programs at Marquette University.

PHD STUDENTS	Ethan Corr, PhD Dissertation Director	2025-present
	Dissertation Topic: <i>TBD</i>	
	Expected Graduation Date: Spring 2028	
	Mahrokh Najaf, PhD Dissertation Director	2023-present
	Dissertation Topic: <i>Performance Guarantees and Efficient Optimization of Implicit Neural Representations for Inverse Problems in Computational Imaging</i>	
	Expected Graduation Date: Spring 2027	
	Megan Murphy (née Lantz), PhD Dissertation Director	2021-present
	Dissertation Topic: <i>Enhancing Deep Learning Image Reconstruction in Breast Computed Tomography with Task-Driven Training and Physics-Informed Models.</i>	
	Expected Graduation Date: Summer 2026	
MASTERS STUDENTS	Ethan Corr, Master’s Essay Director	2024-2025
	Essay Topic: <i>Convex Duality and Uniqueness of Shallow ReLU Networks Trained with Weight Decay Regularization</i>	
	Graduation Date: Spring 2025	
	Henri Medierios Dos Reis, Master’s Essay Director	2023-2024
	Essay Topic: <i>Model-based Image Reconstruction in Computed Tomography: From Iterative To Deep Learning Approaches</i>	
	Graduation Date: May 2024	
	Jessica Pomplun, Master’s Essay Director	2021-2022
	Essay Topic: <i>Constrained Principal Component Analysis with Applications to Matrix Completion</i>	
	Graduation Date: May 2022	
SERVICE ON DISSERTATION COMMITTEES	• Saba Hedari, Marquette University, Comp. Sci. PhD student	2023-present
	• Aryal Milan, Marquette University, Comp. Sci. PhD student	2023-2025
	• Emily Corcoran, Marquette University, CMPS PhD student	2022-2025
	• Jessi Adikorley, Marquette University, CMPS PhD student	2022-2025

Teaching

TEACHING EXPERIENCE	Instructor – Marquette University	Aug 2020 – present
	Served as sole instructor for all courses below. Designed and graded exams, quizzes, in-class activities. For graduate level courses, additionally designed and graded all homework assignments.	
	Differential Equations (4 credits)	Spring 2026
	Theory of Optimization (Graduate level, 3 credits)	Spring 2025
	Seminar: Advanced Optimization in CT (Graduate level, 3 credits)	
	Multivariable Calculus (4 credits)	Fall 2025
	Applied Linear Algebra (Graduate level, 3 credits)	
	Applied Linear Algebra (Graduate level, 3 credits)	Fall 2024
	Calculus II (4 credits)	
	Applied Linear Algebra (Graduate level, 3 credits)	Fall 2023
	Theory of Optimization (Graduate level, 3 credits)	Spring 2023
	Differential Equations (4 credits)	
	Applied Linear Algebra (Graduate level, 3 credits)	Fall 2022
	Differential Equations for Biomedical and Civil Engineers (4 credits)	
	Mathematics of Medical Imaging (Graduate level, 3 credits)	Spring 2022
	Differential Equations (4 credits)	
	Applied Linear Algebra (Graduate level, 3 credits)	Fall 2021
	Differential Equations (4 credits)	
	Theory of Optimization (Graduate level, 3 credits)	Spring 2021
	Differential Equations (4 credits)	
	Differential Equations for Biomedical and Civil Engineers (4 credits)	Fall 2020
	Guest Lectures	
	University of Chicago	Aug 2019
	Course: Machine Learning for Biomedical Informatics	
	• Gave one three-hour lecture introducing deep learning for biomedical image analysis and reconstruction.	
	University of Michigan	Nov 2017
	Course: Matrix Methods for Signal Processing, Data Analysis and Machine Learning.	
	• Gave two lectures in a graduate-level matrix methods course for engineers on the topic of low-rank matrix completion.	
	Teaching Assistant – University of Iowa, Mathematics Department	
	Responsible for leading discussion sections, grading homeworks and exams.	
	Calculus II	Fall 2013
	Multivariable Calculus for Engineers	Spring 2011
	Honors Calculus II	Fall 2010
	Calculus I	Spring 2010
	Calculus I for Biology Students	Fall 2009
	Research Experience for Undergraduates Mentor – U. Iowa	Summer 2011
	Supervisor: Palle Jorgensen, Ph.D.	
	• Led four upper-level undergraduates on an image processing research project.	

Service

ADMIN AND SERVICE

Marquette University, Department of Mathematical and Statistical Sciences

- Graduate Admissions Liaison 2026
- Graduate Committee, member 2022-present
- Graduate Comprehensive Exam Committee, chair 2023-2025
- Faculty Search Committee [ML/AI/Data Science hire], member 2024
- Faculty Search Committee [Stats/Data Science hire], member 2023
- Department Colloquium Coordinator 2021-2022
- Department Website Coordinator 2020-2021
- Undergraduate Academic Advising 2022-present

University of Chicago

- Inverse Problems and Imaging Seminar Series, co-organizer 2019-2020

University of Iowa

- Heartland Talks Liaison 2011-2012
- Student Mathematics Seminar, co-organizer 2011-2012

List of Publications

JOURNAL PUBLICATIONS

1. M. Najaf, **G. Ongie**. "Sampling Theory for Super-Resolution with Implicit Neural Representations." Accepted for publication in SIAM Journal on Imaging Science. To appear 2026.
2. S. Parkinson, **G. Ongie**, R. Willett. "ReLU Neural Networks with Linear Layers are Biased Towards Single-and Multi-index Models." SIAM Journal on Mathematics of Data Science, Vol 7, Issue 3, 1021-1052. July 2025.
3. D. Gilton, **G. Ongie**, R. Willett. "Deep Equilibrium Models for Inverse Problems in Imaging." IEEE Transactions on Computational Imaging, Vol 7, 1123-1133, October 2021.
4. D. Gilton, **G. Ongie**, R. Willett. "Model Adaptation for Inverse Problems in Imaging." IEEE Transactions on Computational Imaging, Vol 7, 661-674, July 2021.
5. E.Y. Sidky, J.P. Phillips, W. Zhou, **G. Ongie**, J. Cruz-Bastida, I.S. Reiser, M.A. Anastasio, X. Pan. "A signal detection model for quantifying over-regularization in non-linear image reconstruction." Medical Physics. 48 (10), 6312-632, January 2021.
6. **G. Ongie**, D. L. Pimentel-Alarcon, L. Balzano, R. Willett, and R. Nowak. "Tensor Methods for Nonlinear Matrix Completion." SIAM Journal on Mathematics of Data Science, 3(1), 253-279, January 2021.
7. **G. Ongie**, A. Jalal, C. Metzler, R. Baraniuk, A. Dimakis, R. Willett. "Deep Learning Techniques for Inverse Problems in Imaging." IEEE Journal on Selected Areas in Information Theory. May 2020.
8. **G. Ongie**^{*}, D. Gilton^{*}, R. Willett. "Neumann Networks for Linear Inverse Problems in Imaging." IEEE Transactions on Computational Imaging, 6, 328-343, 2019.
9. A. Eftekhari, **G. Ongie**, L. Balzano, M. Wakin. "Streaming Principal Component Analysis from Incomplete Data." Journal of Machine Learning Research, 20(86), 1-62, 2019.

10. **G. Ongie** and M. Jacob. "Convex Recovery of Continuous Domain Piecewise Constant Images from Non-Uniform Fourier Samples." IEEE Transactions on Signal Processing, 66(1), 236-250, 2018.
11. **G. Ongie** and M. Jacob. "A Fast Algorithm for Convolutional Structured Low-Rank Matrix Recovery." IEEE Transactions on Computational Imaging, 3(4), 535-550. 2017.
12. **G. Ongie** and M. Jacob. "Off-the-grid Recovery of Piecewise Constant Images from Few Fourier Samples." SIAM Journal of Imaging Sciences, 9(3), 1004–1041. 2016.
13. **G. Ongie** and M. Jacob. "Recovery of Discontinuous Signals Using Group Sparse Higher Degree Total Variation." Signal Processing Letters, 22(9), 1414-1418. 2015.
14. Y. Moshin, **G. Ongie**, and M. Jacob, "Iterative Shrinkage Algorithm for Patch Smoothness Regularized Medical Image Recovery." IEEE Transactions on Medical Imaging. 2015.
15. **G. Ongie***, Y. Hu*, S. Ramani, M. Jacob. "Generalized Higher Degree Total Variation." IEEE Transactions on Image Processing, 23(6), 2423-2435. 2014. **equal authorship*

LONG-FORM
PEER-REVIEWED
CONFERENCE
PUBLICATIONS

16. C. Zeno, H. Manor, **G. Ongie**, N. Weinberger, T. Michaeli, D. Soudry. "When Diffusion Models Memorize: Inductive Biases in Probability Flow of Minimum-Norm Shallow Neural Nets." International Conference on Machine Learning (ICML). July 2025.
17. S. Parkinson, **G. Ongie**, R. Willett, O. Shamir, N. Srebro. "Depth Separation in Norm-Bounded Infinite-Width Neural Networks." Conference on Learning Theory (COLT). 2024.
18. C. Zeno, **G. Ongie**, Y. Blumenfeld, N. Weinberger, D. Soudry. "How do Minimum-Norm Shallow Denoisers Look in Function Space?" Neural Information Processing Systems (NeurIPS). 2023.
19. M.S. Nacson, R. Mulayoff, **G. Ongie**, T. Michaeli, D. Soudry. "The Implicit Bias of Minima Stability in Multivariate Shallow ReLU Networks." International Conference on Representation Learning (ICLR). 2023.
20. **G. Ongie**, R. Willett, D. Soudry, N. Srebro. "A Function Space View of Bounded Norm Infinite-width ReLU Nets: The Multivariate Case." International Conference on Representation Learning (ICLR), 2020.
21. **G. Ongie**, R. Willett, R. Nowak, L. Balzano. "Algebraic Variety Models for High-Rank Matrix Completion." International Conference on Machine Learning (ICML). Sydney, Australia. 2017.

SHORT-FORM
PEER-REVIEWED
CONFERENCE
PROCEEDINGS

22. M. Najaf, **G. Ongie**. "Memory-efficient optimization of implicit neural representations for CT reconstruction." 9th International Conference on Image Formation in X-ray Computed Tomography. June 2026. (Accepted, to appear)
23. **G. Ongie**, M. Murphy, E.Y. Sidky, I.S. Reiser, X. Pan. "Enhancing Signal Detectability in Plug-and-Play CT Reconstruction." SPIE Medical Imaging. Feb 2026.
24. M. Najaf, **G. Ongie**. "Accelerated Optimization of Implicit Neural Representations for CT Reconstruction." IEEE International Symposium on Biomedical Imaging (ISBI). April 2025.

25. **G. Ongie**, M. Murphy, E.Y. Sidky, I.S. Reiser, X. Pan. "Direct optimization of signal detection metrics in learning-based CT image restoration." SPIE Medical Imaging. Feb 2025.
26. M. Najaf, **G. Ongie**. "Towards a Sampling Theory for Implicit Neural Representations." Asilomar Conference on Signals, Systems, and Computers, 2024.
27. **G. Ongie**. "Continuous domain reconstruction in CT imaging with coordinate-based neural networks." International Conference on Image Formation in X-ray Computed Tomography, 2024.
28. **G. Ongie**, B. Chen, Z. Zhang, D. Xia, E.Y. Sidky, X. Pan. "Simultaneous spectral calibration and image reconstruction in spectral CT with a sum of deltas spectrum model." International Conference on Image Formation in X-ray Computed Tomography, 2024.
29. M. Murphy, **G. Ongie**, E.Y. Sidky, I.S. Reiser, X. Pan. "Investigation of Different Model Observers for Including Signal-Detectability in the Training of CNNs for CT Image Reconstruction." SPIE Medical Imaging, 2024.
30. M. Murphy, **G. Ongie**. "Learning-Based Material Decomposition in Dual Energy CT Using an Unrolled Estimator." IEEE International Symposium on Biomedical Imaging, 2023.
31. **G. Ongie**, E.Y. Sidky, I.S. Reiser, X. Pan. "Evaluation of deep learning-based CT reconstruction with a signal-Laplacian model observer." International Conference on Image Formation in X-Ray Computed Tomography, 2022.
32. **G. Ongie**, E.Y. Sidky, I.S. Reiser, X. Pan. "Optimizing model observer performance in learning-based CT reconstruction." SPIE Medical Imaging, 2022.
33. D. Gilton, **G. Ongie**, R. Willett. "Model adaptation for inverse problems in biomedical imaging." IEEE International Symposium on Biomedical Imaging, 2021.
34. J.P. Phillips, E.Y. Sidky, **G. Ongie**, W. Zhou, J. Cruz-Bastida, I.S. Reiser, M.A. Anastasio, X. Pan. "A hybrid channelized Hotelling observer for estimating the ideal linear observer for total-variation-based image reconstruction." SPIE Medical Imaging, 2021.
35. **G. Ongie**, E. Sidky, I. Reiser, X. Pan. "Supervised Learning of Model Observers for Assessment of CT Image Reconstruction Algorithms." SPIE Medical Imaging, 2020.
36. D. Gilton, **G. Ongie**, R. Willett. "Learned Patch-based Regularization for Inverse Problems in Imaging." IEEE International Workshop on Computational Advances in Multi-Sensor Adaptive Processing (CAMSAP). 2019.
37. D. Gilton, **G. Ongie**, R. Willett. "Learning to Regularize with Neumann Networks." IEEE Data Science Workshop. Minneapolis, Minnesota. 2019.
38. **G. Ongie**, D. Hong, D. Zhang, L. Balzano. "Online Estimation of Coherent Subspaces with Adaptive Sampling." IEEE Statistical Signal Processing Workshop. Freiburg, Germany. 2018.
39. **G. Ongie**, N. Murthy, L. Balzano, J. Fessler. "A Memory-efficient Algorithm for Large-scale Sparsity Regularized Image Reconstruction." The Fifth International Conference on Image Formation in X-Ray Computed Tomography. Salt Lake City, Utah. 2018.

40. **G. Ongie**, D. Hong, D. Zhang, L. Balzano. "Enhanced Online Subspace Estimation via Adaptive Sensing." Asilomar Conference on Signals, Systems, and Computers. Pacific Grove, CA. 2017.
41. D.L. Pimentel-Alarcon, **G. Ongie**, L. Balzano, R. Willett, R. Nowak. "Low Algebraic Dimension Matrix Completion." Allerton Conference on Communication, Control, and Computing. Urbana-Champaign, IL. 2017.
42. **G. Ongie**, S. Dewangan, J. Fessler, L. Balzano. "Online Dynamic MRI Reconstruction via Robust Subspace Tracking." IEEE Global Conference on Signal and Information Processing (GlobalSIP). Montreal, Canada. 2017.
43. **G. Ongie**, J. Shi, & J. Fessler. "Efficient Computation of Regularized Field Map Estimates in 3D." IEEE International Symposium on Biomedical Imaging (ISBI). Melbourne, Australia. 2017.
44. **G. Ongie**, S. Biswas, & M. Jacob. "Structured Low-rank Recovery of Piecewise Constant Signals with Performance Guarantees." IEEE International Conference on Image Processing (ICIP). Phoenix, AZ. 2016.
45. A. Balachandrasekaran, **G. Ongie**, & M. Jacob. "Accelerated Dynamic MRI Using Structured Low Rank Matrix Completion." IEEE International Conference on Image Processing (ICIP). Phoenix, AZ. 2016.
46. **G. Ongie** and M. Jacob. "A Fast Algorithm for Structured Low-Rank Matrix Recovery with Applications to Undersampled MRI Recovery." IEEE International Symposium on Biomedical Imaging (ISBI). Prague, Czech Republic. 2016.
47. **G. Ongie** and M. Jacob. "Recovery of Piecewise Smooth Images from Few Fourier Samples." Sampling Theory and Applications (SampTA). Washington, D.C. 2015.
48. **G. Ongie** and M. Jacob. "Super-resolution MRI Using Finite Rate of Innovation Curves." IEEE International Symposium on Biomedical Imaging (ISBI). Brooklyn, NY. *Best student paper award winner*.
49. **G. Ongie**, Y. Hu, M. Jacob. "Higher Degree Total Variation for 3-D Image Recovery." International Symposium on Biomedical Imaging (ISBI). Beijing, China. 2014.
50. Y. Moshin, **G. Ongie**, M. Jacob. Accelerated MRI Using Iterative Non-local Shrinkage. Annual Conference of the Engineering in Medicine and Biology Society (EMBC). Chicago, IL. 2014.